

Varad Patil

varadap25@iitk.ac.in | LinkedIn | GitHub | Portfolio | AI Portfolio

Education

Year	Degree / Examination	Institute / Board	CPI / %
2027	M.Tech, AI for Sustainability	Indian Institute of Technology Kanpur	9.25
2025	B.Tech, Computer Engineering	Rajarambapu Institute of Technology	9.70
2021	Class XII (HSC)	Maharashtra State Board	98.67%
2019	Class X (SSC)	Maharashtra State Board	93.60%

Achievements

- Awarded the **Reliance Foundation Postgraduate Scholarship**, granted to 100 postgraduate scholars nationwide across all disciplines (2025)
- Gold Medalist**, graduated Rank 1 in B.Tech Computer Engineering, Rajarambapu Institute of Technology (2025)
- Top 20 Finalist** of 15,000+ participants, Vibe2Ship National Hackathon; **Finalist**, CityLens AI Hackathon, IIT Kanpur (2026)
- Best Performance Award**, Bubble Byte Ventures Pvt Ltd, for consistent delivery across client projects (2023)

Publications

A RAG-Based Legal Chatbot Using Llama2 and Advanced Information Retrieval 2024

First Author, Proceedings of the International Conference on Trends in Computational and Cognitive Sciences (ICTCS 2024), Volume 6, Springer

- Engineered an end-to-end retrieval-augmented generation pipeline with FAISS vector indexing and Llama2-7B served via LangChain, designing corpus chunking, embedding strategies and prompt templates
- Built an evaluation framework measuring retrieval precision, latency and response quality across 500+ test queries
- Presented the work at the International Conference on Trends in Computational and Cognitive Sciences (2024)

M Tech Thesis

Vision-Based PM2.5/PM10 Estimation from a City-Scale CCTV Network (ongoing) 2026

Advisor: Prof. Sachchida Nand Tripathi, Dean of Kotak School of Sustainability, IIT Kanpur

- Constructing a co-located dataset pairing fixed-camera CCTV clips with reference PM2.5/PM10 monitors and CPCB CAAQMS meteorology on a common observation grid
- Building a resumable multi-stage pipeline for ingest, quality control, sharding, feature extraction and dataset assembly, with atomic manifests, hash-guarded caches and an automated test suite
- Extracting physics-grounded features (RMS contrast, dark-channel prior, Koschmieder extinction coefficient and optical flow) alongside frozen DINOv2 ViT embeddings
- Fine-tuning per-camera ResNet-18 regressors on camera imagery, benchmarked against a gradient-boosting ladder that attributes predictive contribution across optics, embeddings and meteorology
- Designing a leakage-controlled evaluation protocol with day-, week- and camera-blocked splits and paired blocked-bootstrap significance testing
- Reimplementing the Wang et al. (2023) CNN-LSTM as an independent comparator to benchmark against published vision-based PM estimation
- Extending the design toward sensor-augmented gap-filling and short-horizon forecasting, targeting camera-assisted augmentation of existing instrumented sites

Professional Experience

Software Development Intern, Altair Engineering, Bangalore Jan–Jun 2025

- Built full-stack MERN client portals for tire manufacturers, enabling engineers to configure, submit and track HPC simulation workflows
- Built a real-time HPC job scheduling and visualisation dashboard surfacing live job status, queue position and simulation results
- Improved frontend performance by 30% through code-splitting, lazy loading and API caching; implemented role-based authentication and automated job-submission pipelines
- Automated simulation and reporting workflows using Python and REST APIs, reducing manual operations by 40%
- Authored technical documentation for client onboarding and developer handover, covering portal workflows, API usage and deployment procedures

Web Development Intern, Bubble Byte Ventures Pvt Ltd, Sangli Sep 2022–Aug 2023

- Delivered 10+ responsive, SEO-optimised client websites, reducing page load times by 30% via asset optimisation and caching
- Automated deployment pipelines with Python and shipped cross-browser, mobile-responsive interfaces
- Designed and maintained client websites and web applications end to end, handling requirement gathering, revisions and post-delivery support

Key Projects

Active-Break Indian Summer Monsoon Forecasting (Course Project: Applications of AI in Sustainability Lab, KSS605)  | PyTorch, TCN, Transformers, XGBoost, ERA5 Reanalysis

- Built a TCN–Transformer meta-ensemble classifying Active, Break and Normal monsoon phases up to 7 days ahead from ERA5 reanalysis data and engineered atmospheric features
- Lifted mean Macro F1 to 0.560 against an XGBoost baseline at 0.521, with the largest gain at the 7-day horizon where Break-class F1 rose from 0.198 to 0.493
- Derived labels from IMD gridded rainfall anomalies using thresholds computed on training years only, and enforced chronological train, validation and test splits to prevent leakage between neighbouring seasons

Cyclone Track Prediction System (Course Project: Fundamentals of Data Science and Machine Intelligence, EE708)  | XGBoost, Gradient Boosting, Scikit-learn, IBTrACS

- Engineered 40+ predictive features from the IBTrACS dataset (motion vectors, intensity gradients and spatial context) to forecast positions at 6h, 12h, 24h, 48h and 72h horizons
- Achieved **14.47 km RMSE at 6h** and 78.77 km at 24h; built an evaluation pipeline with geodesic error metrics and skill-score analysis benchmarked against a persistence baseline
- Added persistence-extrapolated position features as a physics anchor so models learn track deviations rather than absolute coordinates, and split train and test at storm level to prevent track leakage

CivicPulse: Multi-Agent Civic Issue Platform (Self Project)  | Gemini 2.5-flash, Next.js 16, React 19, TypeScript, Firestore, Docker, Cloud Run

- Designed an orchestrated multi-agent pipeline: multimodal triage extracting category, severity and hazards; automated complaint routing to municipal authorities with priority and SLA; and before/after repair verification by image comparison
- Applied a minimal-AI architecture, reserving inference for perception and judgement while keeping deduplication, geospatial hotspot prediction and routing deterministic
- Built a full-stack platform with Firebase Authentication (Google and anonymous), server-verified ID tokens, Cloud Storage behind a private media proxy, and an interactive Leaflet map with marker clustering and multi-layer filtering
- Containerised with Docker and deployed on Google Cloud Run via Cloud Build, operating entirely within cloud free tiers
- Implemented a community verification workflow with status lifecycle tracking, contributor tiers, achievement badges and a points leaderboard

CityLens: Urban Infrastructure Monitoring (Self Project)   | YOLO11, YOLO26, PyTorch, Ultralytics, SAHI, DDP

- Trained YOLO11m on the MBDD2025 UAV dataset (14.5K images, 57.6K annotations, 5 classes) reaching **0.918 test mAP@0.5**, using multi-label iterative stratification and rare-class oversampling for a 10:1 class imbalance, trained via DDP on dual T4 GPUs
- Built an illegal-billboard detector at 0.813 test mAP@0.5 through a five-variant ablation with a from-scratch AP@0.5 metric (IoU matching, precision-recall interpolation); validation/test gap tracking identified added model capacity as overfitting
- Evaluated SAHI tiled inference (320px slices, 20% overlap, NMS merge) as a data-representation alternative to added model parameters, concluding that dataset scale rather than architecture was the binding constraint

Air Quality Early Warning System (Self Project)  | Gradient Boosting, Scikit-learn, Flask, Python

- Built next-day AQI forecasting for Indian cities using temporal feature engineering and lag-based predictors with CPCB health-advisory thresholding, served via a deployable REST API for public health alerting
- Trained paired regression and six-class CPCB bucket classification models on 108K station-day records across 110 monitoring stations, reaching an R^2 of 0.92 and 20.99 AQI mean absolute error on a held-out test set
- Served recursive seven-day forecasts through a Flask REST API with startup caching, backed by a dashboard rendering AQI trends, leading pollutant values and health advisories

Resume Insight: LLM Resume Evaluation Engine (Self Project)  | Next.js 16, React 19, Gemini API, Clerk, Supabase

- Built semantic resume to job-description matching with ATS scoring, keyword alignment and line-level improvement suggestions, deployed publicly with persistent analysis history
- Engineered a Gemini-backed analysis service over PDF-extracted resume text, exposing separate endpoints for job-match scoring, overall ATS analytics and section-level rewriting with per-request token and error logging
- Implemented tiered subscriptions on Clerk authentication and Supabase Postgres, enforcing per-feature monthly quotas, upload size limits and automatic usage-period resets

Delhi Jal Board WhatsApp CRM (Self Project)  | Node.js, Express.js, MongoDB, Mongoose, WhatsApp Cloud API

- Built a grievance-management backend integrating the WhatsApp Cloud API into an automated chatbot workflow for citizen complaint registration, status tracking and resolution across water, sewer and revenue categories
- Designed REST APIs and MongoDB schemas for complaint lifecycle management, enabling citizens to raise and monitor grievances without a web portal or call centre
- Supported concurrent multi-user sessions through per-phone session isolation, request queuing for rapid messages, webhook deduplication and atomic document updates, verified by a dedicated concurrency test suite

Snikrz: Sneaker E-Commerce Platform (Self Project)  | Next.js 16, React 19, MongoDB, Mongoose, Stripe

- Built a full-stack sneaker store with JWT authentication, role-based access control, cart, order tracking and admin inventory dashboards
- Integrated the Stripe payment gateway with secure transaction processing and inventory management
- Implemented SEO and delivery optimisations including JSON-LD structured data, a dynamically generated sitemap, Open Graph metadata and PWA manifest support

Spotify 2.O: Music Streaming Platform (Self Project) 🎧 | Next.js, TypeScript, Supabase, Stripe, WebSockets

- Built a personalised streaming application with Supabase authentication and storage, playlist management, audio playback and Stripe-based premium subscription billing
- Built real-time music rooms over a WebSocket service with shareable room codes, synchronised song selection and persisted chat for multiple simultaneous listeners
- Implemented Stripe subscription checkout with trial periods and promotion codes, synchronising products, prices, customers and subscription status into Supabase through signed webhooks

Technical Skills

Languages: Python, JavaScript, TypeScript

GenAI and LLM: RAG Systems, LangChain, FAISS, Llama2, Gemini, OpenAI API, HuggingFace, Prompt Engineering, Multi-Agent Systems, Multimodal Inference, NLP

Deep Learning: PyTorch, TensorFlow, Transformers, TCN, CNNs, Object Detection (YOLO, Ultralytics)

Machine Learning: Scikit-learn, XGBoost, Gradient Boosting, Reinforcement Learning, Time-Series Forecasting, Feature Engineering, Sampling and Imbalance Handling, Ablation and Baseline Analysis, Skill-Score and Generalisation Gap Analysis

Scientific Data: NumPy, Pandas, Matplotlib, ERA5 Reanalysis, IBTrACS, CPCB and CAAQMS Air Quality Data

Frontend: React.js, Next.js, Redux, Tailwind CSS, Framer Motion, HTML5, CSS3

Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, JWT Authentication

Databases: MongoDB, PostgreSQL, MySQL, Firestore, Prisma, Mongoose

Cloud and DevOps: GCP (Cloud Run, Firestore, Firebase), Docker, Vercel, Git, GitHub

Geospatial: Geospatial Libraries (Leaflet, OpenStreetMap), Spatial Clustering and Hotspot Analysis

Mobile and Payments: Flutter, Google Play Store Deployment, Stripe, Razorpay

CS Fundamentals: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

Positions of Responsibility

- **Teaching Assistant**, Mathematics and Computation Using Python, IIT Kanpur, under Prof. Mahendra Kumar Verma (2026)
- **Community Lead**, Google Developer Groups on Campus: led a technical community of 45+ members, organised 5+ workshops on web development and Generative AI, mentored 20+ students (2024–25)

Certifications

- Supervised Machine Learning: Regression and Classification, DeepLearning.AI and Stanford University (2025)
- Generative AI with Large Language Models, DeepLearning.AI and AWS (2024)
- Programming Foundations with JavaScript, HTML and CSS, Duke University (2020)

Extra-Curricular Activities

- **Independent Application Developer:** publish and maintain mobile applications on the Google Play Store under an independent developer account, handling design, development, release and post-launch support end to end (2026)
- **Freelance Web Development:** delivered a production website for a paying fitness-events client, covering design, development, payment integration and deployment (2024)